

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Proposed Solution

ShopEZ — MERN Stack E-commerce Application

Phase: Project Design Phase

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Classification: Project Documentation — For Academic Submission

Proposed Solution

1. System Context Diagram

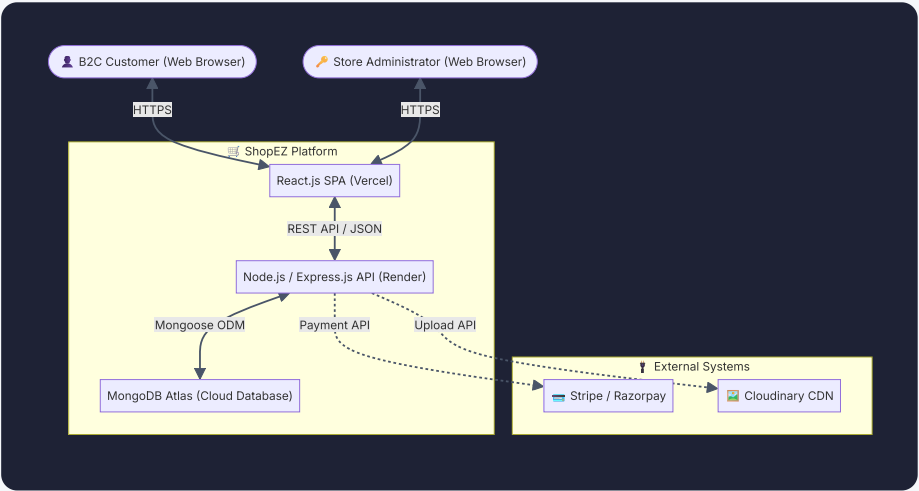


FIGURE 8.1 — C4 MODEL LEVEL 1: SYSTEM CONTEXT DIAGRAM

2. Solution Parameters

S.No.	Parameter	Detailed Description
1	Problem Statement	Customers require a secure, fast, and intuitive platform to discover and purchase items online. Administrators require a centralised, efficient tool to manage product inventory, monitor sales performance, and fulfil orders. The core gap is the absence of a unified, modern, customisable e-commerce solution accessible without prohibitive SaaS subscription costs.
2	Solution Description	A fully responsive Single Page Application (SPA) built on the MERN stack. The platform provides: (i) a customer-facing storefront with robust product discovery, persistent cart, and guided checkout; and (ii) an administrator panel with inventory management, order fulfilment, Cloudinary-powered image uploads, and sales analytics dashboards. Payment processing is delegated to PCI-DSS compliant gateways (Stripe / Razorpay).
3	Novelty / Uniqueness	Decoupled architecture with independently deployable frontend and backend; MongoDB-backed cross-device cart persistence; Cloudinary dynamic image transformations on CDN delivery; stateless JWT authentication enabling horizontal API scaling.
4	Social Impact / Customer Satisfaction	Delivers a reliable, transparent purchasing process building consumer trust through secure payment integration and clear order tracking. Empowers small business operators with accessible analytics, enabling data-

S.No.	Parameter	Detailed Description
		driven inventory decisions and improved customer service.
5	Business Model (Revenue Model)	Primary: direct sales margin on products. Secondary: premium product placement / featured listing slots; percentage-based commission for future multi-vendor configuration; shipping and handling fee collection on high-volume orders.

3. High-Level Feature Summary

Feature Category	Key Capabilities
Authentication & Security	JWT + Bcrypt; HTTP-only cookie sessions; RBAC (User / Admin)
Product Discovery	Full-text search; Category, Price, Rating filters; Sorting options
Shopping Experience	Persistent cross-device cart; Real-time total calculation; Stock validation
Checkout & Payment	3-step wizard; Stripe / Razorpay integration; Invoice generation
Order Management	Order history (customer); Fulfilment status tracking (admin)
Product Reviews	Verified purchase reviews; Aggregate rating recalculation
Admin Operations	Product CRUD; Category management; Cloudinary image upload
Admin Analytics	Revenue charts; Order volume metrics; User growth indicators

